

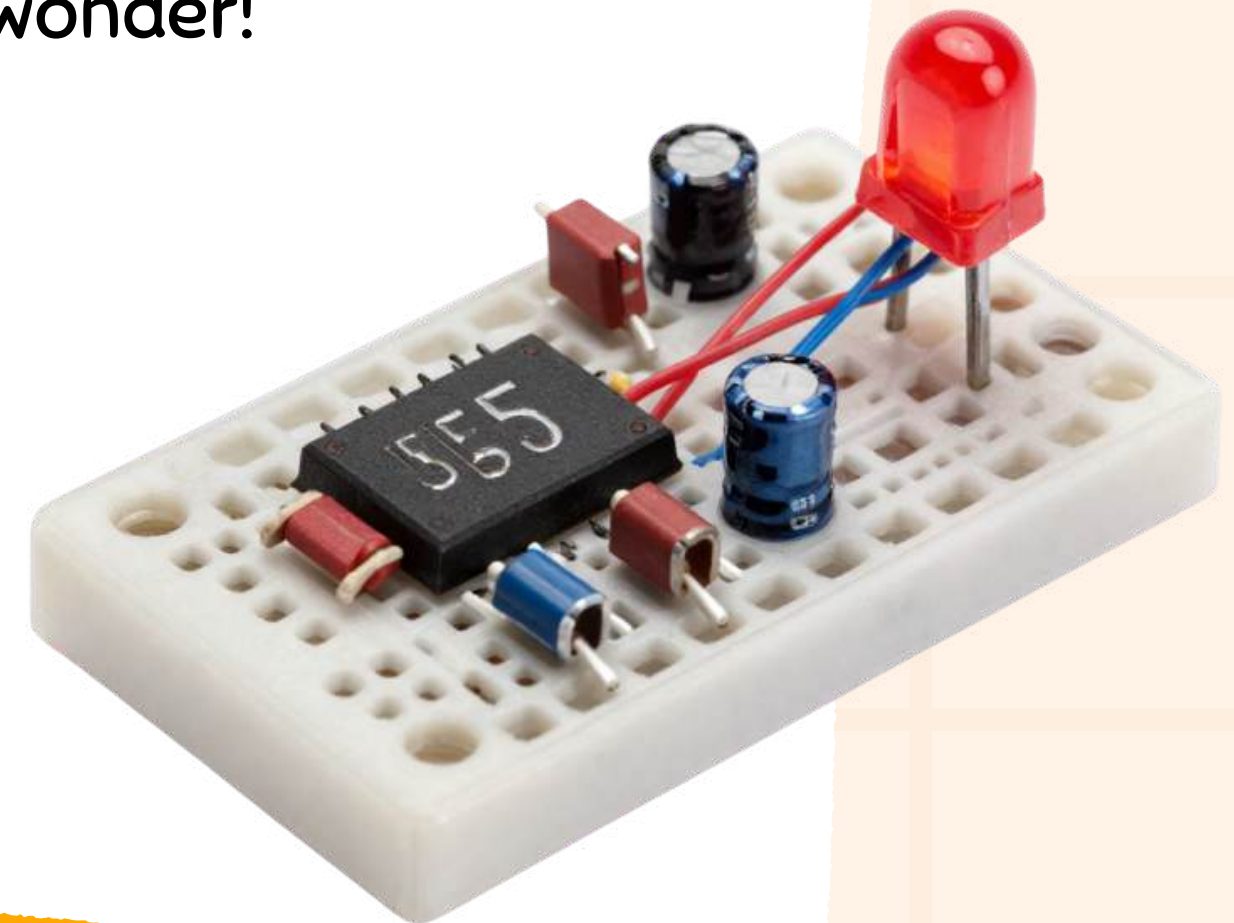
555 Timer IC – Astable Oscillator





Look! A Blinking LED Circuit

Can you guess how this LED keeps blinking on and off by itself? Watch closely and wonder!



What Do You Know About Circuits?

Let's think together! Answer these fun questions.
There are no wrong answers – just share what you know or guess!



What makes a light bulb glow?

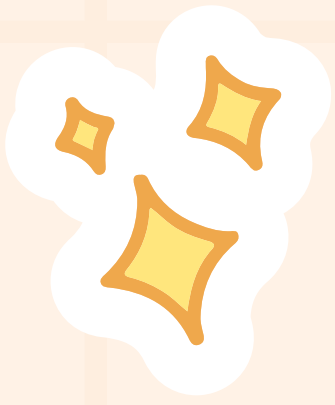


Have you seen something blink on its own?



What do you think an IC is?





What Is an IC?

An IC is like a school tiffin box (dabba). It packs many tiny parts inside, just like your tiffin box holds many food items!



Meet the 555 Timer IC

The 555 Timer is a small chip that helps devices keep time. It costs only about Rs. 5 to Rs. 10! We use it to make things blink and buzz!



Tiny but powerful chip



Helps devices keep perfect time



Makes LEDs blink and buzzers beep





Astable Mode – The Dholak Player

Astable means never stable. It keeps switching on and off, just like a dholak player at a wedding – a steady rhythm that never stops!





What Does Astable Mode Do?

It makes an LED blink on and off again and again. No need for us to press any button!
It's automatic blinking!



Blinks ON and OFF forever



No button pressing needed



Works automatically!

The 8 Pins of the 555 Timer

This IC has 8 pins – tiny legs to connect it to a circuit. Each pin has a special job to help the timer work properly!



Pin 1: GND, Pin 2: Trigger



Pin 3: Output, Pin 4: Reset, Pin 5: Control



Pin 6: Threshold, Pin 7: Discharge, Pin 8: VCC





Key Pins to Remember

The 555 Timer IC has 8 pins, but let's focus on the 3 most important ones that make our LED blink!



Pin 1: GND (Ground) – the negative side of power



Pin 8: VCC – the positive power supply (+)

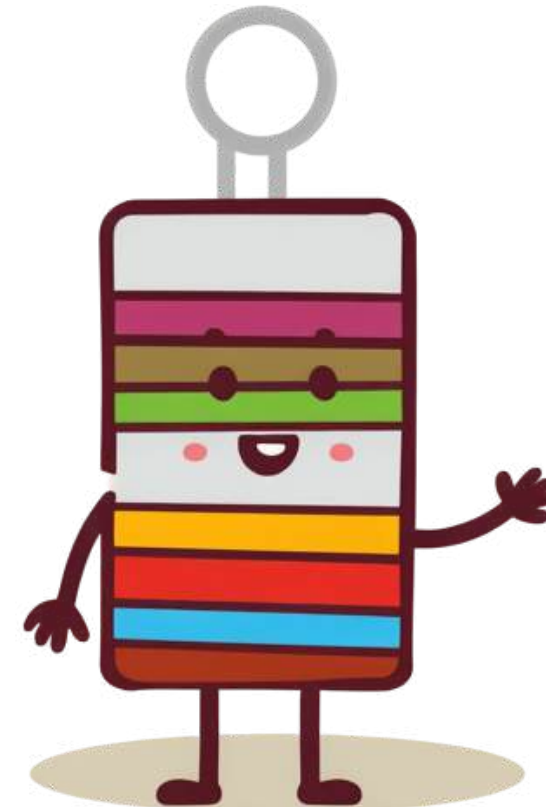


Pin 3: Output – where the blinking signal comes out!



What Are Resistors?

Resistors are like speed breakers on a road. They slow down the flow of electricity, just like speed breakers slow down cars!





What Is a Capacitor?

A capacitor is like a (clay pot). It fills with electric charge and then empties it out – just like how a Clay pot fills with water and pours it out!



How RA, RB, and C Control Blink Speed

The resistors RA and RB and the capacitor C decide how fast the LED blinks. Change their values to control the blinking speed!



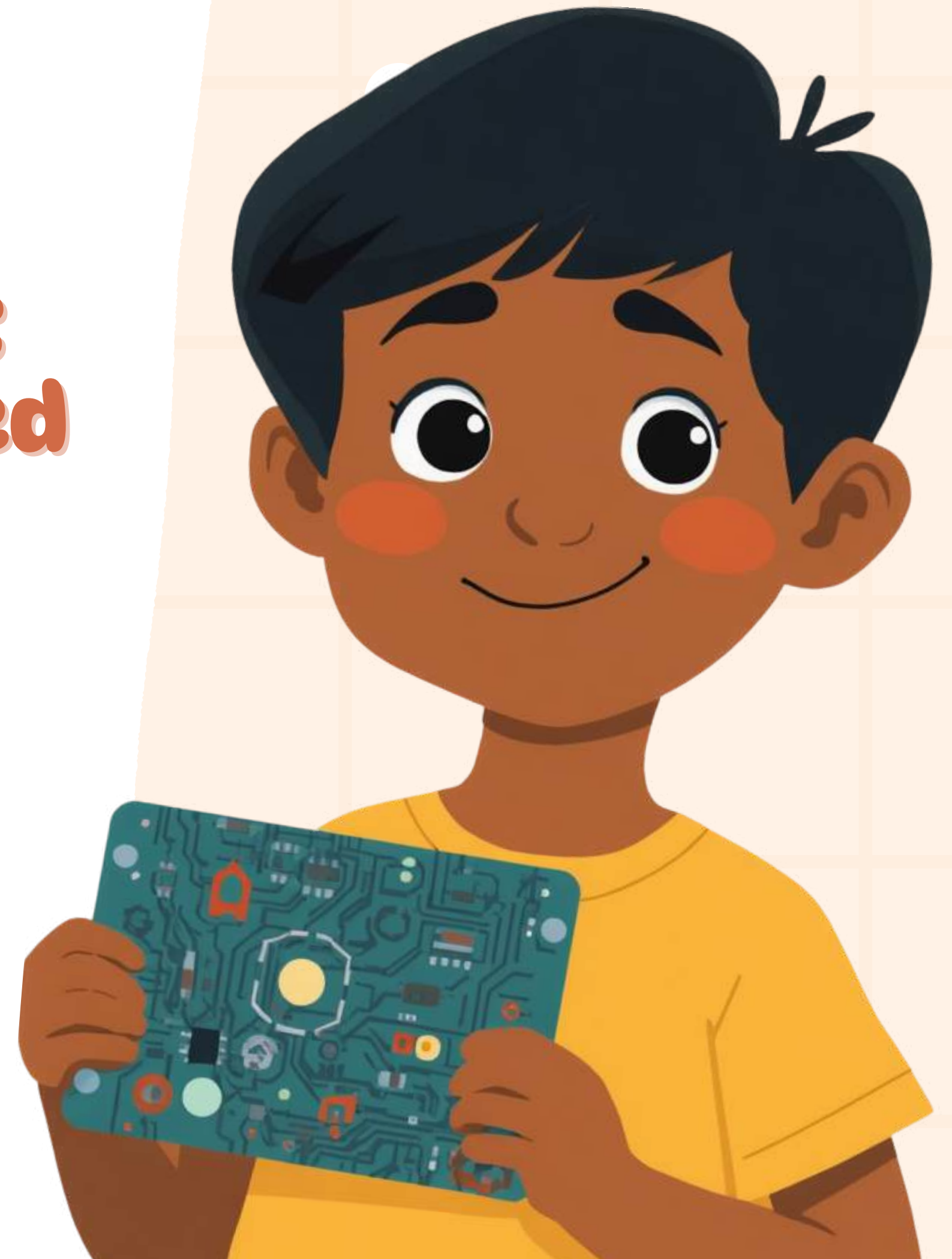
Bigger resistors or capacitor = slower blinking



Smaller resistors or capacitor = faster blinking



You control the blink rhythm!



Blink Speed Made Simple

Blinking speed is like how fast a traffic signal changes. Some lights blink slowly, some blink quickly!



Slow blink = waiting at traffic lights



Fast blink = bicycle tail light flashing quickly



You can control the speed with resistors and capacitors!



Indian Real-World Applications

The 555 Timer IC is used in many things you see every day in India! Let's explore where blinking circuits help us.



Railway crossing signals and traffic lights



Bicycle tail lights that flash for safety



Diwali lights and doorbell chimes





Why Learn About 555 Timer?

We use the 555 Timer to make many things work automatically. Understanding it helps us create cool gadgets!



Build fun blinking circuits



Create your own gadgets



Become a young inventor

Let's Recap What We Learned

Great job today! Here are the key ideas we explored about the 555 Timer IC and how it makes things blink automatically.



IC = tiffin box with many parts; 555 Timer has 8 pins



Astable mode = dholak rhythm that never stops



Resistors = speed breakers; Capacitor = clay pot



Quick Quiz Time!

Question: What does an IC do?
Think about what we learned today and choose the right answer!



a) Packs many parts inside



b) Makes food



c) Plays music



Quick Quiz Time! (2)

Question: "Astable mode means the LED..."
Think about what we learned! Take a guess!



a) Stays ON always



b) Blinks ON and OFF forever



c) Never turns ON



Quick Quiz Time! (3)

What is a resistor like? Think about what slows things down!



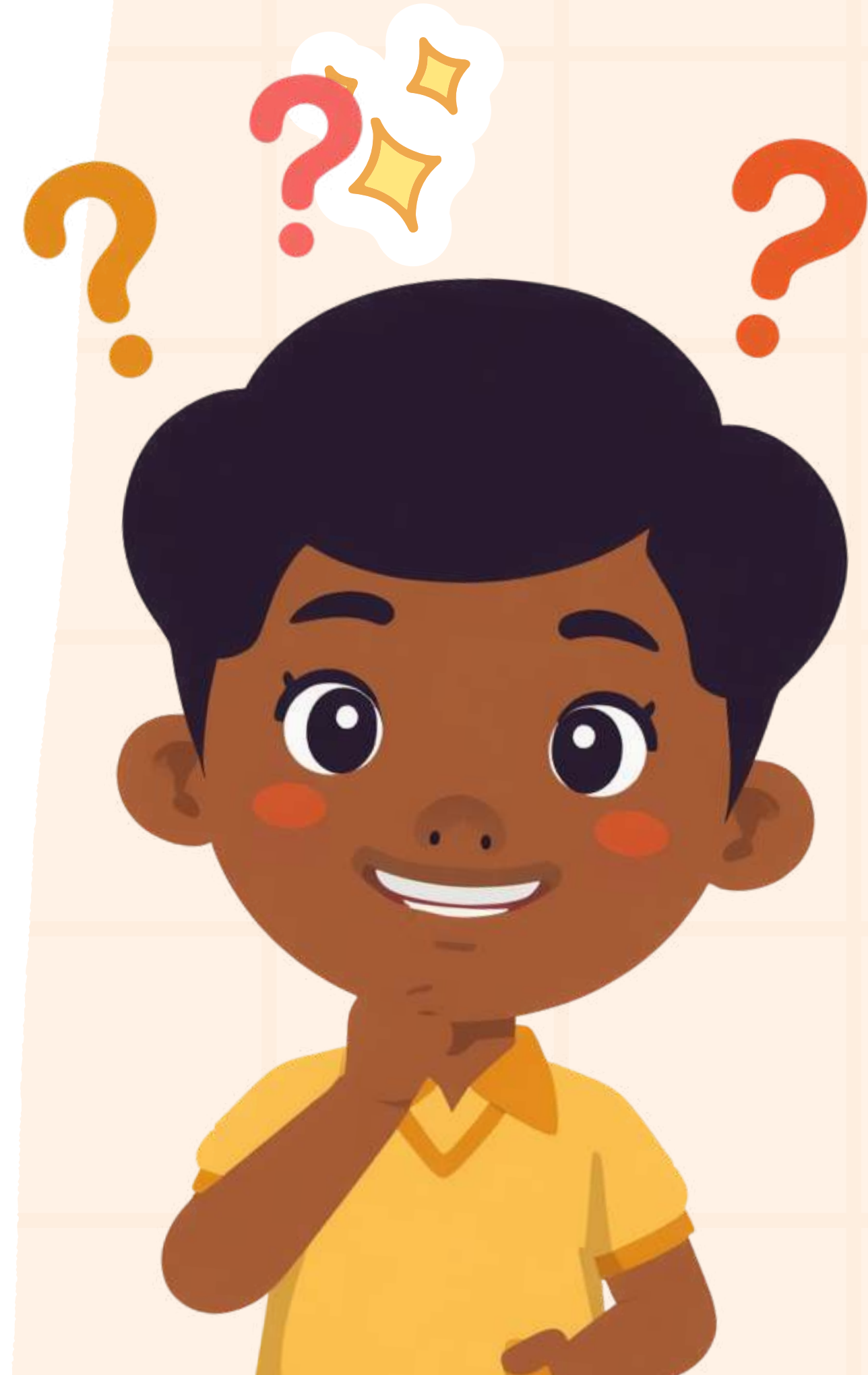
a) Speed breaker



b) Water pot



c) Drum



Homework for Next Time

Here are some fun activities to try at home before our next session. Share what you discover with your family!



Draw your own 555 Timer IC with all 8 pins labeled



Find an example of blinking lights at home



Tell a family member what you learned today



Thank You!

Great job today! Keep exploring
and asking questions. See you
next session!





Questions?

Any questions? We're happy to help!
Reach out to Planrs Academy anytime.

